



Introduction

The damped harmonic oscillator is a cornerstone of classical mechanics, describing systems where a restoring force and a damping force act on an object. This report delves into the behavior of the damped harmonic oscillator under various parameters, analyzing the effects on position, velocity, phase space, and energy. By numerically solving the differential equations governing the system, we generate and examine graphs to compare theoretical expectations with simulated results.

Theoretical Background

$\frac{d^2x}{dt^2} + 2\beta\frac{dx}{dt} + \omega_0^2 = F_0\cos(\omega t)$ This equation describes the dynamics of a system. x is usually considered as the displacement function of the system, while β represents the damping and ω_0 the natural frequency. On the right side there is a diving force $F_0\cos(\omega t)$ as an external force. To get the equation in the form $m\ddot{x} + a\dot{x} + kx = F_0\cos(\omega t)$ each term is substituted appropriately:

$$m\ddot{x} = \frac{d^2x}{dt^2}$$

$$a\dot{x} = 2\beta\frac{dx}{dt}$$

$$kx = \omega_0^2 x$$

With these terms, the given equation can be rewritten as follows:

$$m\ddot{x} + a\dot{x} + kx = F_0\cos(\omega t) \text{ where:}$$

m: mass

F₀: amplitude of the dicing force

$\dot{x}$: velocity

a: damping coefficient

ω : angular frequency of the diving force

$\ddot{x}$: acceleration

k: spring constant

x: displacement

Methodology

Reasons for Selection of Variables:

The variables studied in the project (mass, spring constant, damping coefficient, driving force amplitude, angular frequency, initial velocity and time step) have been selected in a wide range in order to understand the dynamic behavior of the system and to study the energy transformations. This choice was made in order to observe and analyze in detail the effects of different parameters on the system. Each variable represents the main physical properties of the system and the influence of external forces.

Define the parameters

- Mass **m(kg)** : [5.0, 10.0, 15.0]
- Spring constant **k(N/m)** : [10.0, 20.0, 30.0]
- Force amplitude **F₀(N)** : [60.0, 80.0, 100.0]
- Frequency **ω (rad/s)** : [10.0, 15.0, 20.0]
- Initial position **x₀(m)** : [100.0, 200.0, 300.0]
- Initial velocity **v₀(m/s)** : [5.0, 10.0, 15.0]
- Total time **T(s)** : different for differet graphs
- Time step **dt(s)** : 0.01

Tables and Incremental Data:

The data in the tables are organized in increments to better understand and compare the effects of different parameters on the system. This arrangement allows to systematically observe the effects of parameter variations on the behavior of the system and to evaluate the consistency of the results. For a clearer understanding of the relationships between the parameters, different values of each variable are used to analyze the responses of the system.

QR Code:

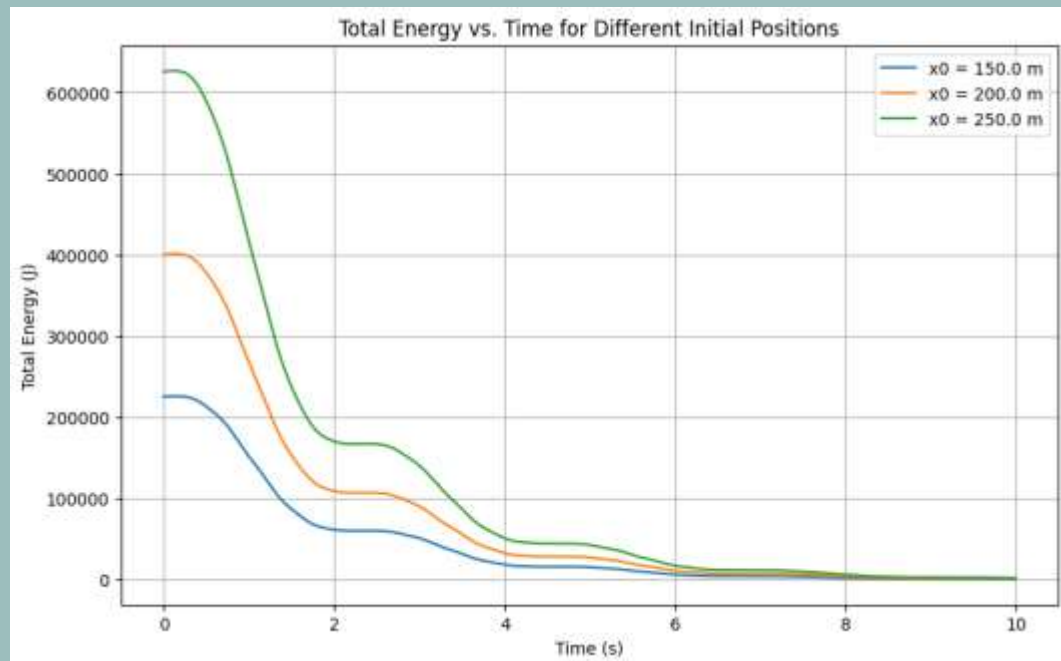
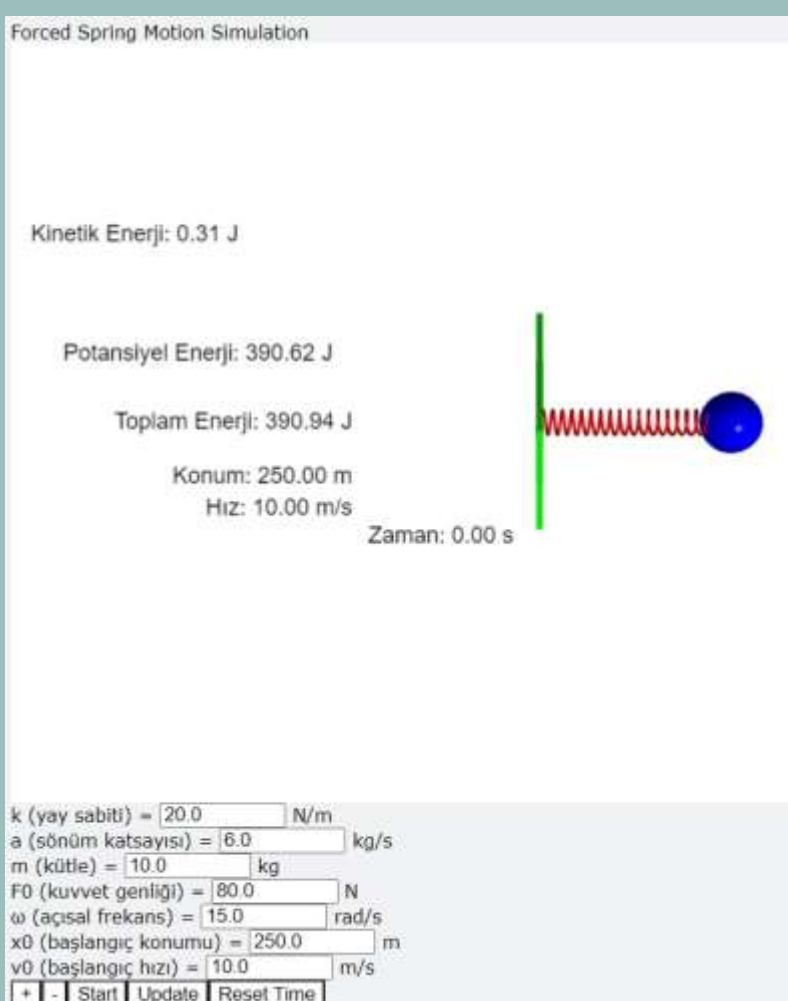
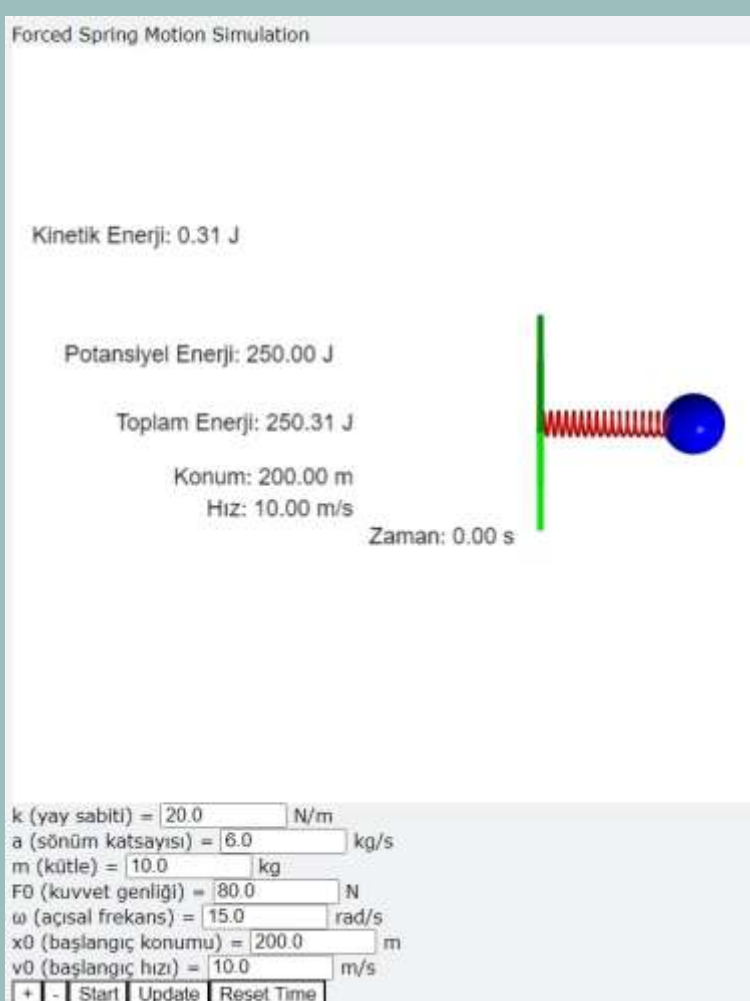
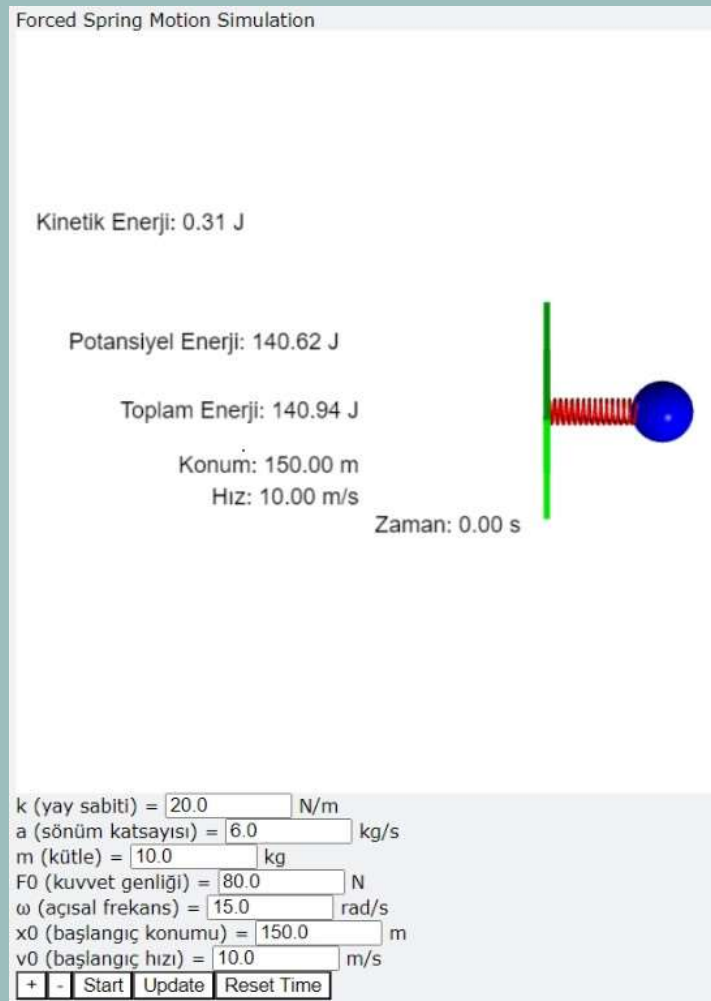
The QR code on the poster provides access to all the codes and the detailed report of the project. This provides access to all the data and documentation for those who want to learn more in-depth about the project and examine the methods used. By scanning the QR code, comprehensive information about the algorithms, calculations and analysis used in the project can be accessed.

Numerical Solution Method:

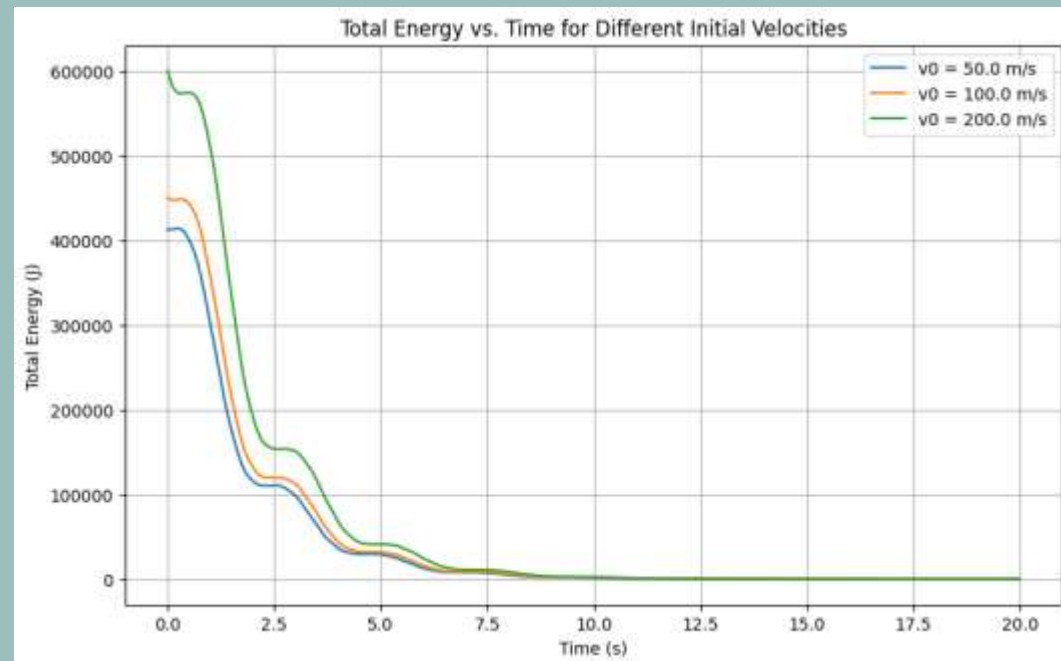
Euler's method was used for the numerical solution of the equation studied in the study. The Euler method is a method used in the numerical solution of differential equations. With this method, the given equation is solved numerically using time steps and the time dependent behavior of the system is analyzed.

Graphs, Analysis and Observations

Energy graph for different initial positions

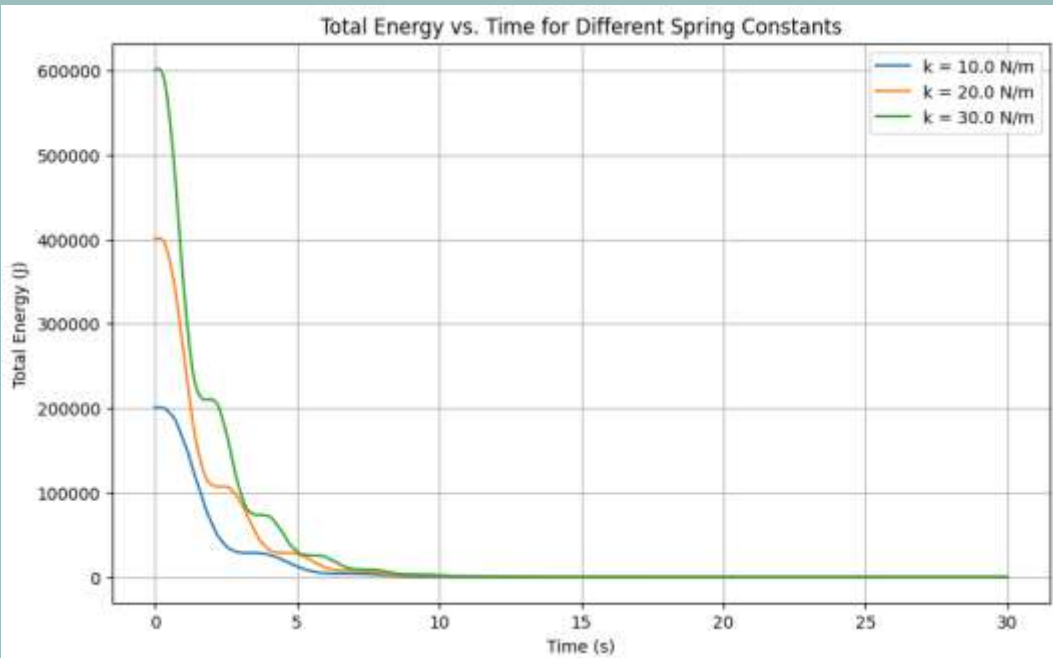


The initial position x_0 influences the system's initial potential energy, with higher x_0 leading to greater initial total energy. During oscillations, total energy fluctuates due to the exchange between kinetic and potential energy, with more pronounced fluctuations for larger initial displacements. Despite these fluctuations, total energy remains approximately constant, demonstrating energy conservation, with deviations attributed to damping or numerical errors. Over time, total energy stabilizes as energy input from external forces balances with energy dissipation. Systems with higher initial positions exhibit higher total energy and more pronounced fluctuations, with larger displacements leading to more rapid energy exchanges between kinetic and potential energy.

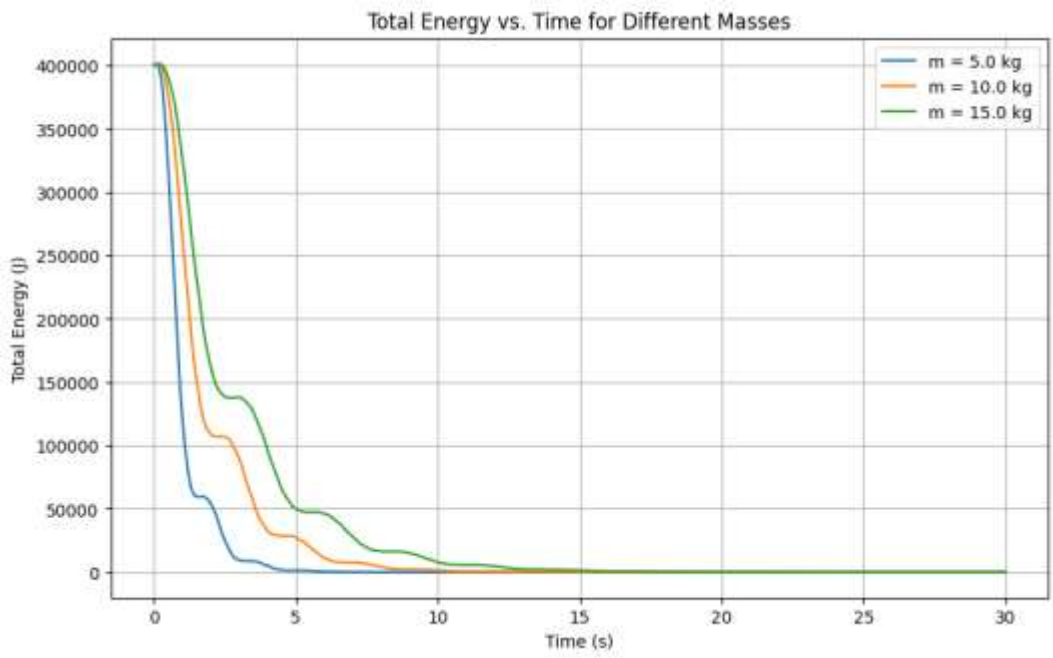


The initial velocity v_0 determines the system's initial kinetic energy, with higher v_0 resulting in greater initial total energy. As the system oscillates, total energy fluctuates due to the exchange between kinetic and potential energy, with higher velocities causing more rapid oscillations and larger fluctuations. Despite these fluctuations, total energy remains approximately constant over time, showing energy conservation, with deviations due to damping or numerical errors. Over time, total energy stabilizes as energy input from external forces balances with energy dissipation. Systems with higher initial velocities exhibit higher total energy and more pronounced fluctuations, with more rapid energy exchanges between kinetic and potential energy.

Energy Graphs



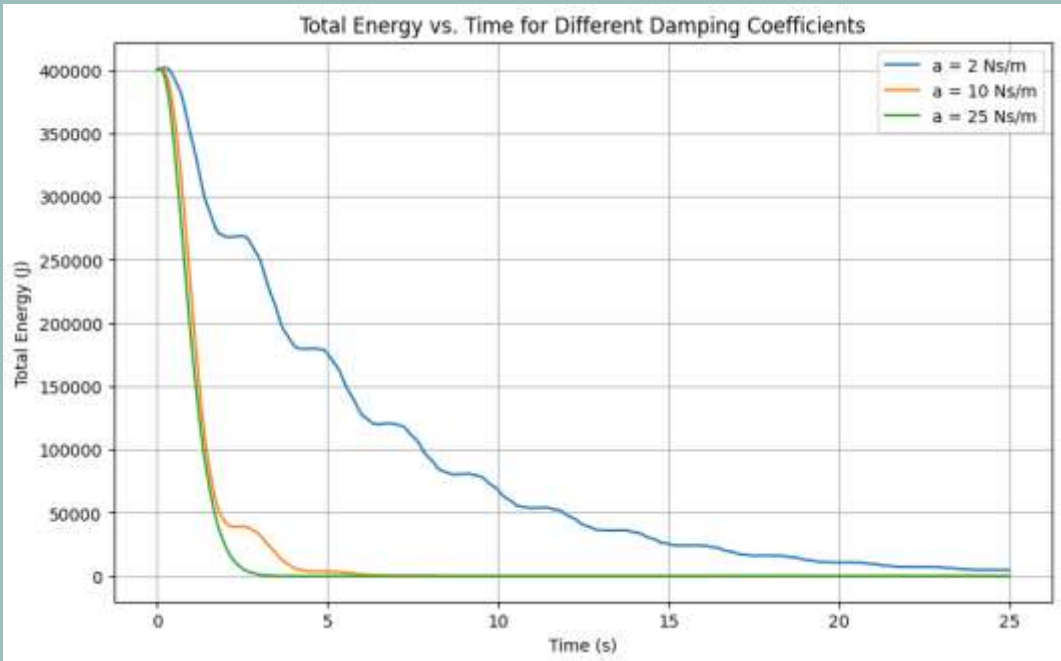
The total energy $E(t)$ of a damped harmonic oscillator decreases over time due to the damping effect, represented by the exponential term $e^{-\beta t}$ in the displacement equation, where $\beta = \frac{a}{2m}$. As the spring constant k increases, the system's stiffness and initial potential energy $U = \frac{1}{2}kx^2$ also increase, leading to higher initial total energy $E(0)$. The rate of energy dissipation is influenced by the damping coefficient a and is independent of k , although higher k values start with higher initial energies and exhibit more rapid oscillations in energy. Despite this, the damping effect ensures that all systems eventually dissipate their energy. Therefore, higher spring constants result in higher initial energies and more noticeable oscillatory behavior in the energy decay process, but ultimately, all systems will dissipate their energy over time due to damping.



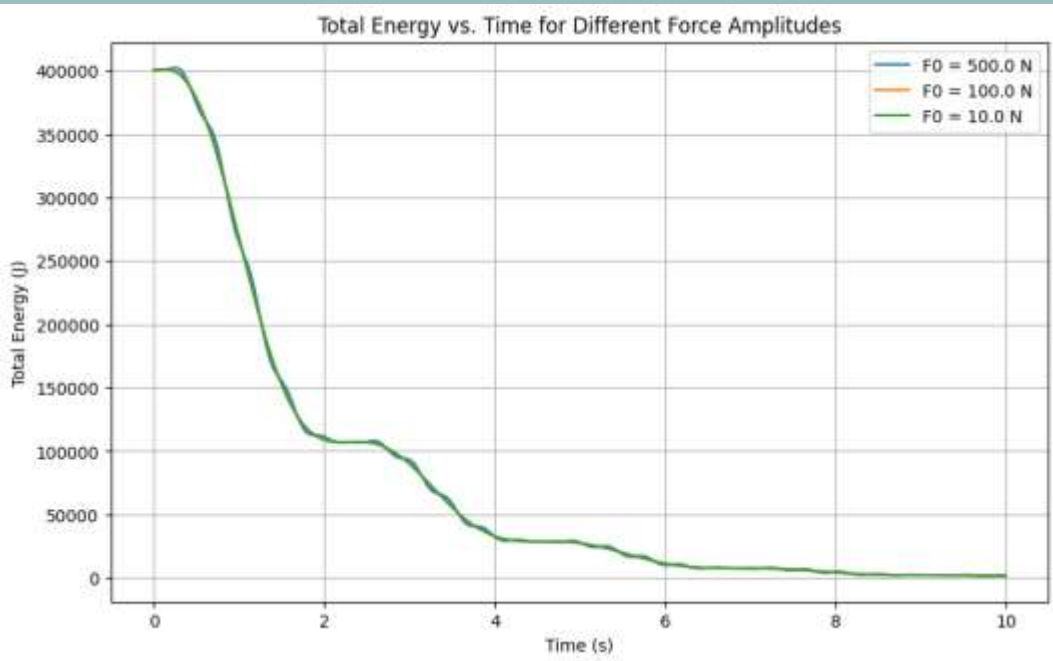
The total mechanical energy $E(t) = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$ of a damped harmonic oscillator combines kinetic K and potential U energy. Larger mass increases kinetic energy for the same velocity, while potential energy depends on the spring constant and displacement. Energy dissipation, influenced by the damping coefficient a , follows $e^{-\beta t}$ with $\beta = \frac{a}{2m}$. Higher mass leads to lower angular

frequency $\omega_o = \sqrt{\frac{k}{m}}$, slower oscillations, and slower

energy dissipation. Thus, systems with greater mass have higher initial energies, slower oscillations, and more gradual energy decay, highlighting the relationship between mass, oscillation frequency, and energy dissipation.



The total energy $E(t)$ of a damped harmonic oscillator decreases over time due to damping, with the decay rate increasing with the damping coefficient a and following $E(t) \approx E_0 e^{-\beta t}$, where $\beta = \frac{a}{2m}$. The initial energy E_0 is constant across different a values, set by x_0 and v_0 . Higher damping coefficients lead to faster energy dissipation: $a = 2.0\text{Ns/m}$ shows slow decay (underdamped), $a = 10.0\text{Ns/m}$ shows moderate decay, and $a = 25.0\text{Ns/m}$ shows rapid decay (overdamped). Higher a values lead to quicker energy reduction to zero, highlighting the role of damping in energy dissipation.



The initial total energy E_0 is directly proportional to the force amplitude F_0 and is determined by initial conditions x_0 and v_0 . Higher force amplitudes result in higher initial total energy, contributing to the system's potential energy $PE = \frac{1}{2}kx^2$. Ideally, in the absence of external input or dissipation, total energy remains constant, but with damping and external forcing, energy fluctuates over time. These fluctuations are more pronounced with higher F_0 , as the system responds more strongly to the force. Over time, total energy stabilizes as damping balances external energy input. For $F_0 = 40\text{N}$, fluctuations are small; for $F_0 = 80\text{N}$, they are moderate; and for $F_0 = 120\text{N}$, they are significant, indicating a highly responsive system.

Frame from animation

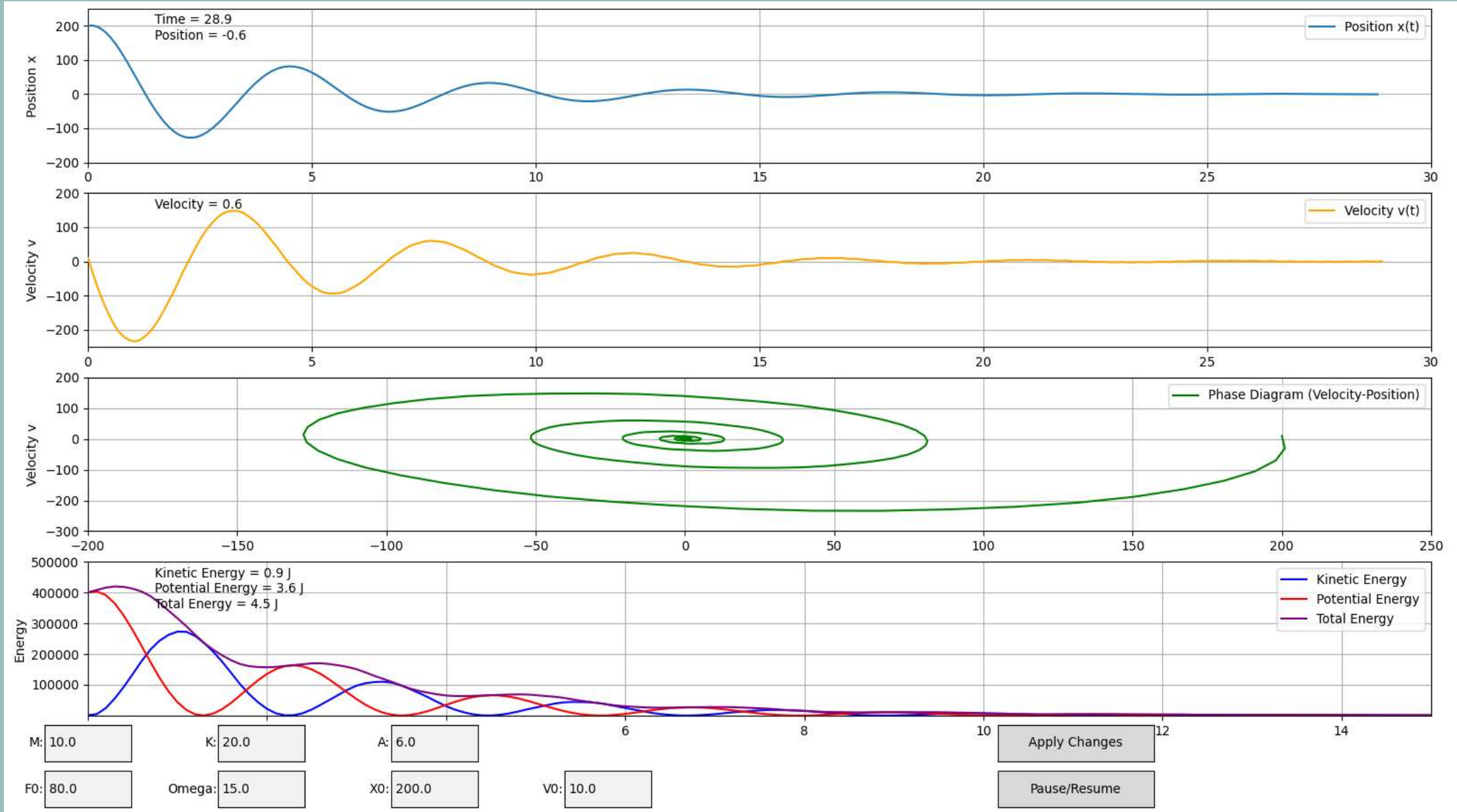


Table of Effects of Variables on Each Other

Independent variables	Substitution(x)	velocity(V)	Total Energy (E)	Kinetic Energy (K)	Potential Energy (U)
Mass (m)	↑	↓	↑	↑	↑
Damping Coefficient (a)	↓	↓	↓	↓	↓
Spring Constant (k)	↑	↑	↑	↑	↑
Coercive Force Amplitude (F ₀)	↑	↑	↑	↑	↑
Angular Frequency of Coercive Force (ω)	↑	↑	↑	↑	↑
Starting Position (x ₀)	↑	—	↑	—	↑
Initial Speed (v ₀)	—	↑	↑	↑	—

Conclusion and Evaluation

The results of the project show how the system behaves with various parameters and the effects of these parameters on the system dynamics. The simulations and analyses performed within the scope of the study revealed the following important findings:

Effect of Initial Position:

- Higher initial positions cause the system to have more potential energy at the beginning and this leads to higher total energy values.
- As the initial position increased, the energy changes became more pronounced, indicating that the system oscillated more as it moved away from the equilibrium point.

Effect of Damping Coefficient:

- High damping coefficients resulted in faster energy dissipation and slower oscillations and lower velocities.
- Low damping coefficients, on the other hand, allowed the energy to remain in the system longer and caused faster oscillations and higher speeds.
- Damping showed that mechanical energy is not conserved and decreases with time, and energy is dissipated in the form of heat or other forms.

Effect of the Angular Frequency of the Forcing Force:

- Higher angular frequencies caused a faster change of energy between the kinetic and potential energy components.
- The energy fluctuations of the system were more pronounced at higher angular frequencies, indicating that the system responds faster to external forces.
- However, due to the damping effect, the effect of angular frequency was less pronounced in the overall energy plot when energy loss was dominant.

Energy Conservation and Fluctuations:

- The system showed energy fluctuations in practice due to damping and external forces, even when energy conservation was theoretically valid.
- Over time, the total energy of the system tended to stabilize at a certain value, indicating that the energy input by external forces was balanced by the energy loss by damping.

These findings show that the project has achieved its goal and clearly demonstrates how the dynamic behavior of the system changes depending on the specified parameters. The simulation results are in line with theoretical expectations and provide valuable insights to understand how such systems will behave in real-world applications.

